DAVID ADEOGUN

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years in ML and 7+ years of software engineering, specializing in model development, training, debugging, optimization, and evaluation across applied research and production systems. Designed and deployed end-to-end ML pipelines spanning computer vision, natural language processing, time-series analysis, and robotics perception, from synthetic data generation to production deployment and monitoring. Experienced building scalable ML infrastructure on AWS (SageMaker, ECS, S3) and Kubernetes, with a demonstrated ability to bridge cutting-edge research and production-grade engineering.

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, LSTMs, CNNs, Vision Transformers, AdaBoost, YOLO, XGBoost, LightGBM, Random Forests, Neural Networks, ONNX Runtime, TensorFlow Serving, Hugging Face Transformers, Stable Diffusion, Reinforcement Learning

Experimentation & Evaluation: MLflow, Weights & Biases, Experiment Tracking, Model Registry, Hyperparameter Tuning, Optuna, Ray Tune, A/B Evaluation, Benchmark Design, RMSE/Precision/Recall/F1 Metrics, Cross-Validation, Feature Importance Analysis, SHAP

ML Infrastructure & Ops: AWS (SageMaker, ECS, ECR, S3, Lambda), Kubernetes, Docker, Apache Airflow, Kubeflow, CI/CD for ML Pipelines, GitHub Actions, Automated Rollbacks, Model Versioning, DVC, Feature Stores, Production Monitoring, Prometheus, Grafana

Programming & Data: Python, C++, SQL, MATLAB, NumPy, Pandas, Apache Kafka, MQTT, Spark, dbt, PostgreSQL, Redis, REST APIs, FastAPI

Perception & Robotics: OpenCV, PCL, ROS/ROS2, LiDAR Processing, 3D Point Cloud Segmentation, Sensor Fusion, SLAM, Computer Vision, Depth Estimation, Object Detection

INDUSTRY EXPERIENCE

Machine Learning Engineer | Mode Robots — Oct 2023 – Present

- Designed and trained temporal ML models including LSTMs and AdaBoost classifiers for predictive maintenance, achieving earlier fault detection with a 5,000+ cycle lead time, less than 5% false alarm rate, and RUL prediction RMSE under 2,000 cycles across time-series datasets with 150,000+ labeled cycles.
- Engineered feature extraction and selection pipelines across structured time-series and sensor data, applying dimensionality reduction techniques to improve model accuracy and reduce training time across high-dimensional IIoT telemetry datasets.
- Developed synthetic data generation pipelines with realistic noise injection and multiple degradation patterns — linear, exponential, and logarithmic — using NumPy and Pandas, creating controlled training and benchmarking environments for model robustness evaluation.
- Designed and evaluated CNN and Vision Transformer architectures for image-based perception tasks, benchmarking classification accuracy, inference speed, and model generalization across remote sensing and satellite imagery datasets.
- Fine-tuned and evaluated large language models for domain-specific reasoning tasks, designing schema-driven evaluation pipelines using AWS Bedrock to benchmark groundedness, decision-logic, and response quality across frontier models.
- Assessed and debugged AI-generated solutions across the full ML lifecycle, identifying failure modes in model training and optimization pipelines using PyTorch, scikit-learn, and ONNX Runtime to improve convergence and generalization.
- Benchmarked temporal vs. non-temporal architectures, conducting detailed error analysis on sequential data processing capabilities to inform production model selection and ML system design tradeoffs.
- Applied hyperparameter optimization using Optuna and Ray Tune across model training runs, improving model performance metrics including RMSE, precision, recall, and F1 scores across multiple production ML systems.
- Implemented model versioning, experiment tracking, and model registry management using MLflow and Weights & Biases, ensuring full reproducibility across training runs and deployment stages.
- Containerized ML models using Docker and deployed scalable inference endpoints on AWS ECS and Kubernetes, optimizing TensorFlow Serving and ONNX Runtime for production latency and throughput.

Cloud Platform Engineer | Natterbase — Feb 2022 – Sep 2023

- Designed and deployed cloud-native infrastructure on AWS to support client-facing consumer applications, ensuring high availability, fault tolerance, and seamless scaling during peak usage periods.
- Built and managed CI/CD pipelines to automate application deployment workflows, reducing release cycle time and enabling consistent, zero-downtime deployments across development, staging, and production environments.

- Configured and maintained EKS-based containerized workloads for client application services, orchestrating multi-container deployments using Kubernetes to optimize resource utilization and application reliability.
- Implemented observability and monitoring solutions using Prometheus and Grafana to track application health, infrastructure performance, and client-facing service uptime across production environments.

Software Engineer | Williams Sonoma Inc Jun 2020 – Jan 2022

- Developed and maintained vendor-facing web applications enabling real-time tracking of customer order fulfillment across warehouse locations, improving supply chain visibility and reducing order status inquiry resolution time.
- Built and optimized backend services supporting order lifecycle management between distribution centers, ensuring data accuracy and reliability for high-volume retail operations across vendor and warehouse touchpoints.

Software Developer | Finance Life Technologies Aug 2015 - May 2020

- Developed and maintained core modules for an enterprise lending platform supporting customer onboarding, loan approval workflows, and loan servicing, enabling financial access for underserved and unbanked populations.
- Built and integrated backend services for automated loan processing pipelines, improving approval turnaround time and ensuring regulatory compliance across customer lifecycle stages from application to repayment.

RESEARCH

Kennesaw State University – *Digital Automation Research Lab*

Advisor: Professor David Guerra-Zubiaga

- Trained and compared LSTM time-series model against AdaBoost baseline classifier for predictive maintenance, achieving earlier fault detection (5,000+ cycle lead time), <5% false alarm rate, and RUL prediction RMSE <2,000 cycles.
- Designed digital twin of ABB IRB 1200 robotic cell in Siemens Tecnomatix Process Simulate, validating IIoT-to-ML data pipeline architecture for Industry 4.0 predictive maintenance evaluation.
- Engineered synthetic IIoT telemetry generation pipeline producing 150,000+ labeled cycles with realistic noise injection and three degradation patterns (linear, exponential, logarithmic) for model training and benchmarking.
- Demonstrated that temporal modeling (LSTM on 500-cycle sequential windows) captures gradual degradation trends missed by snapshot-based approaches, informing ML system design tradeoffs for time-series tasks.

Georgia Southwestern State University – *Intelligent Systems Research*

Advisor: Professor Alexander Yemelyanov

- Architected AI-driven decision-support framework combining LLM reasoning, structured decision logic, and cognitive-behavioral problem-solving to produce explainable recommendations for risk-sensitive scenarios.
- Embedded Azure OpenAI (GPT-4o) as the core personalization engine, designing schema-driven user-profile pipelines to generate grounded, user-specific trade-offs and alternative rankings.
- Productionized and evaluated AI systems on Google Cloud with guardrails, telemetry-based quality monitoring, and cross-platform QA, delivering end-to-end architecture documentation.

PUBLICATIONS

[1] Yemelyanov, A., Adeogun, D. "AI-Powered Cognitive-Behavioral Decision Support for Risk Reduction in Insurance Coverage Selection." AHFE 2026. (Submitted)

PREPRINTS

[1] Adeogun, D., Vu, K., Alam, W., Quansah, E. "Digital Twin–Driven Predictive Maintenance for Pick-and-Place Robotic Cells Using IIoT Telemetry and Machine Learning." ASME IMECE 2026. (To Be Submitted) https://doi.org/10.13140/RG.2.2.15447.53926

[2] Adeogun, D., Sandoval, S., Wulfekuhle, V., Vu, K., Kouwenhoven, F. "Improving Efficiency of Deep-Learning based Vision Transformer for Remote Sensing for Satellite Images." ResearchGate, January 2026. https://doi.org/10.13140/RG.2.2.13764.49282

EDUCATION

M.S. Computer Science | Georgia Southwestern State University

- *Thesis: Design and Implementation of a User-Centered AI-Assisted Decision-Support Framework Under Risk and Uncertainty*

M.S. Intelligent Robotic Systems | Kennesaw State University

- *Relevant Coursework: Machine Learning, Machine Vision, Perception/Navigation/Path Planning, Robot Simulation, 3D Point Cloud Manipulation*

B.S. Software Development | Brigham Young University-Idaho